UN R32 (version 0.2.0)

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1. SCOPE

This Regulation applies to vehicles of category M_1 1/ with regard to their behaviour in a rear-end collision.

[1] As defined in the Consolidated Resolution on the Construction of Vehicles (R.E.3.), document ECE/TRANS/WP.29/78/Rev.**7**, para. 2 - **https://unece.org/transport/vehicle-regulations/wp29/resolutions**

2. DEFINITIONS

2.1. In defining the dimensions the following simplified terminology is used:

2.1.1. a "transverse plane" is a vertical plane perpendicular to the median longitudinal plane of the vehicle;

2.1.2. a "longitudinal plane" is a plane parallel to the median longitudinal plane of the vehicle;

2.1.3. the vehicle's "reference plane" is a plane related to the vehicle and horizontal when the vehicle in the state defined in paragraph 2.2.4. below rests on a horizontal plane.

2.2. For the purposes of this Regulation:

2.2.1. "approval of a vehicle" means the approval of a vehicle type with regard to the behaviour of the structure of the passenger compartment in a rear-end collision;

2.2.2. "vehicle type" means a category of power-driven vehicles which do not differ in such essential respects as:

2.2.2.1. the length and width of the vehicle in so far as they have an effect on the results of the impact test prescribed in this Regulation;

2.2.2.2. the structure, dimensions, lines and material of the part of the vehicle rearward of the transverse plane through the "R" point of the rearmost seat;

2.2.2.3. the lines and inside dimensions of the passenger compartment in so far as they have an effect on the results of the impact test prescribed in this Regulation;

2.2.2.4. the siting of the engine (front, rear, centre).

2.2.3. "passenger compartment" means the space for occupant accommodation, bounded by the roof, floor, side walls, doors, outside glazing, front bulkhead, and the plane of the rear compartment bulkhead or the plane of the rear seat back support;

2.2.4. "unladen kerb weight" means the weight of the vehicle in running order, unoccupied and unladen but complete with fuel, coolant, lubricant, tools and a spare wheel (if provided as standard equipment by the vehicle manufacturer);

2.2.5. "capacity of the fuel tank" means the fuel-tank capacity as specified by the manufacturer.

3. APPLICATION FOR APPROVAL

3.1. The application for approval of a vehicle type with regard to the behaviour of the structure of the passenger compartment in a rear-end collision shall be submitted by the vehicle manufacturer or by his duly accredited representative.

3.2. It shall be accompanied by the undermentioned documents in triplicate and by the following particulars:

3.2.1. a detailed description of the vehicle type with respect to its structure, dimensions, lines and constituent materials;

3.2.2. drawings of the vehicle showing the vehicle type in front, side and rear elevation and design details of the rear part of the structure; and

3.2.3. particulars of the vehicle's unladen kerb weight.

3.3. A vehicle representative of the type to be approved shall be submitted to the technical service responsible for conducting the approval tests.

3.3.1. A vehicle not comprising all the components proper to the type may be accepted for test provided that it can be shown that the absence of the components omitted has no detrimental effect on the results of the tests, so far as the requirements of this Regulation are concerned.

3.3.2. It shall be the responsibility of the applicant for approval to show that acceptance of the variants referred to in paragraph 3.3.1. is compatible with compliance with the requirements of this Regulation.

4. APPROVAL

4.1. If the vehicle submitted for approval pursuant to this Regulation meets the requirements of Paragraphs 5. and 6. below, approval of that vehicle type shall be granted.

4.2. An approval number shall be assigned to each type approved. Its first two digits (at present 00 for the Regulation in its original form) shall indicate the series of amendments incorporating the most recent major technical amendments made to the Regulation at the time of issue of the approval. The same Contracting Party may not assign the same number to another vehicle type within the meaning of paragraph 2.2.2. above.

4.3. Notice of approval or of extension or of refusal or withdrawal of approval or production definitely discontinued of a vehicle type pursuant to this Regulation shall be communicated to the Parties to the Agreement which apply this Regulation by means of a form conforming to the model in annex 1 to this Regulation.

4.4. There shall be affixed, conspicuously and in a readily accessible place specified on the approval form, to every vehicle conforming to a vehicle type approved under this Regulation,

4.4.1. an international approval mark consisting of:

4.4.1.1. a circle surrounding the letter "E" followed by the distinguishing number of the country which has granted approval, 2/ and

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[2] The distinguishing numbers of the Contracting Parties to the 1958 Agreement are reproduced in Annex 3 to the Consolidated Resolution on the Construction of Vehicles (R.E.3), document ECE/TRANS/WP.29/78/Rev.7, Annex 3 - **https://unece.org/transport/vehicle-regulations/wp29/resolutions**

4.4.1.2. the number of this Regulation, followed by the letter "R", a dash and the approval number to the right of the circle prescribed in paragraph 4.4.1.1.

4.5. If the vehicle conforms to a vehicle type approved, under one or more other Regulations annexed to the Agreement, in the country which has granted approval under this Regulation, the symbol prescribed in paragraph 4.4.1.1. need not be repeated; in such a case the additional numbers and symbols of all the Regulations under which approval has been granted in the country which has granted approval under this Regulation shall be placed in vertical columns to the right of the symbol prescribed in paragraph 4.4.1.1.

4.6. The approval mark shall be clearly legible and be indelible.

4.7. The approval mark shall be placed close to or on the vehicle data plate affixed by the manufacturer.

4.8. Annex 2 to this Regulation gives examples of arrangements of approval marks.

5. REQUIREMENTS

5.1. When the vehicle has undergone the test referred to in paragraph 6. below, the displacement referred to in annex 4, paragraph 3, to this Regulation shall not exceed 75 mm.

5.2. After the test, no rigid component in the passenger compartment shall constitute a risk of serious injury to the vehicle's occupants.

5.3. In addition, the side doors of the vehicle shall not open under the effect of the impact.

5.4. Except in the case of a vehicle not having a roof of rigid construction, the opening of a number of doors sufficient to enable all the occupants to emerge shall, after the impact, continue to be possible without the use of tools.

6. TEST

6.1. The vehicle's compliance with the requirements of paragraph 5. above shall be checked by the method set out in annexes 3 and 4 to this Regulation.

7. MODIFICATIONS AND EXTENSION OF APPROVAL OF THE VEHICLE TYPE

7.1. Every modification of the vehicle type shall be notified to the administrative department which approved that vehicle type. The department may then either:

7.1.1. consider that the modifications made are unlikely to have appreciable adverse effects, and that in any case the vehicle still meets the requirements; or

7.1.2. require a further test report from the technical service responsible for conducting the tests.

7.2. Without prejudice to the provisions of paragraph 7.1. above, a variant of the vehicle whose unladen kerb weight is lower than the weight of the vehicle submitted for approval shall not be regarded as a modification of the vehicle type.

7.3. Confirmation or refusal of approval, specifying the alterations shall be communicated by the procedure specified in paragraph 4.3. above to the Parties to the Agreement which apply this Regulation.

7.4. The competent authority issuing the extension of approval shall assign a series number for such an extension and inform thereof the other Parties to the 1958 Agreement applying this Regulation by means of a communication form conforming to the model in annex l to this Regulation.

8. CONFORMITY OF PRODUCTION

8.1. Every vehicle bearing an approval mark as prescribed under this Regulation shall conform to the vehicle type approved, more particularly as regards features affecting the behaviour of the structure of the passenger compartment in a rear-end collision.

8.2. In order to verify conformity as prescribed in paragraph 8.1. above, a sufficient number of serially-produced vehicles bearing the approval mark required by this Regulation shall be subjected to random checks.

8.3. As a general rule, the checks aforesaid shall be confined to the taking of measurements. However, the vehicles shall, if necessary, be subjected to the checks prescribed in paragraph 6. above.

9. PENALTIES FOR NON-CONFORMITY OF PRODUCTION

9.1. The approval granted in respect of a vehicle type pursuant to this Regulation may be withdrawn if the requirement laid down in paragraph 8.1. above is not complied with or if a vehicle has failed to pass the tests prescribed in paragraph 6. above.

9.2. If a Party to the Agreement which applies this Regulation withdraws an approval it has previously granted, it shall forthwith notify the other Parties to the Agreement which apply this Regulation by means of a communication form conforming to the model in annex 1 to this Regulation.

10. PRODUCTION DEFINITELY DISCONTINUED

If the holder of the approval completely ceases to manufacture the type approved in accordance with this Regulation, he shall so inform the authority which granted the approval. Upon receiving the relevant communication that authority shall inform thereof the other Parties to the 1958 Agreement applying this Regulation by means of a communication form conforming to the model in annex 1 to this Regulation.

11. NAMES AND ADDRESSES OF TECHNICAL SERVICES CONDUCTING APPROVAL TESTS, AND OF ADMINISTRATIVE DEPARTMENTS

The Parties to the Agreement which apply this Regulation shall communicate to the Secretariat of the United Nations the names and addresses of the technical services conducting approval tests and of the administrative departments which grant approval and to which forms certifying approval or refusal or withdrawal of approval, issued in the other countries, are to be sent.

ANNEX 1 COMMUNICATION

(Maximum format: A4 (210 x 297 mm))



Issued by: Name of administration

.....

.....

.....

concerning: 2/ APPROVAL GRANTED

APPROVAL EXTENDED

APPROVAL REFUSED

APPROVAL WITHDRAWN

PRODUCTION DEFINITELY DISCONTINUED

of a vehicle type with regard to the behaviour of the structure of the passenger compartment in a rear-end collision, pursuant to Regulation No. 32

Approval No.. ..... Extension No. .....

1/ Distinguishing number of the country which has granted/extended/refused/withdrawn approval (see approval provisions in the Regulation).

2/ Strike out what does not apply.

1. Trade name or mark of the vehicle .....

2. Vehicle type.....

3. Manufacturer's name and address .....

4. If applicable, name and address of manufacturer's representative .....

.....

5. Brief description of the vehicle type as regards its structure, dimensions, lines and constituent materials

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6. Weight of vehicle when tested:

front axle: .....

rear axle:.....

total: .....

7. Vehicle submitted for approval on.....

8. Technical service responsible for conducting approval tests .....

.....

9. Date of test report issued by that service .....

10. Number of test report issued by that service .....

11. Approval granted/extended/refused/withdrawn 2/.....

2/ Strike out what does not apply.

12. Position of approval mark on the vehicle .....

13. Place .....

14. Date.....

15. Signature.....

The list of documents deposited with the Administrative Service which has granted approval is annexed to this communication and may be obtained on request.

ANNEX 2 ARRANGEMENTS OF APPROVAL MARKS

Model A

(See Paragraph 4.4. of this Regulation)

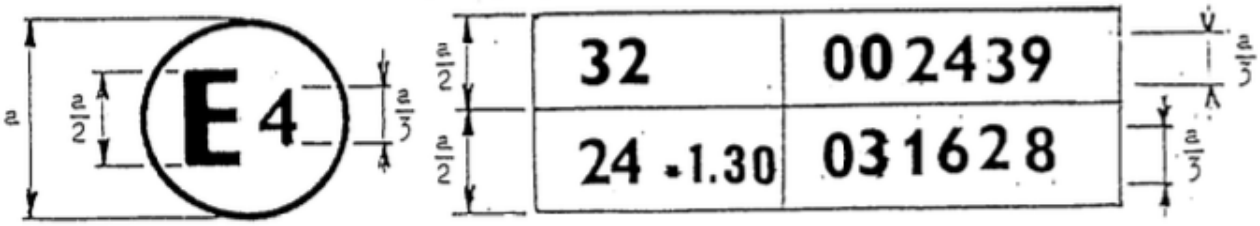


a = 8 mm min.

The above approval mark affixed to a vehicle shows that the vehicle type concerned has with regard to the behaviour of the structure of the passenger compartment in a rear-end collision, been approved in the Netherlands (E 4), under the number 002439. The first two digits of the approval number indicate that the approval was granted in accordance with the requirements of Regulation No. 32 in its original form.

Model B

(See Paragraph 4.5. of this Regulation)



a = 8 mm min.

The above approval mark affixed to a vehicle shows that the vehicle type concerned has been approved in the Netherlands (E 4) pursuant to Regulations Nos. 32 and 24. 1/ (In the case of the latter Regulation the corrected absorption co-efficient is 1.30 m^{1}). The approval numbers indicate that on the dates on which these approvals were granted Regulation No. 32 had not yet been modified and Regulation No. 24 already included the 03 series of amendments.

1/ The second number is given merely as an example.

Annex 3 Procedure for determining the "H" point and the actual torso angle for seating positions in motor vehicles 1

[1] The procedure is described in Addendum 6 of Mutual Resolution No. 1 (M.R.1) (document ECE/TRANS/WP.29/1101/Amend.5); see https://unece.org/transport/vehicle-regulations/wp29/resolutions

Annex 3 Appendix 1 - Description of the three dimensional "H" point machine 1

[1] The procedure is described in Addendum 6 of Mutual Resolution No. 1 (M.R.1) (document ECE/TRANS/WP.29/1101/Amend.5); see https://unece.org/transport/vehicle-regulations/wp29/resolutions

Annex 3 Appendix 2 - Three-dimensional reference system 1

[1] The procedure is described in Addendum 6 of Mutual Resolution No. 1 (M.R.1) (document ECE/TRANS/WP.29/1101/Amend.5); see https://unece.org/transport/vehicle-regulations/wp29/resolutions

Annex 3 Appendix 3 - Reference data concerning seating positions 1

[1] The procedure is described in Addendum 6 of Mutual Resolution No. 1 (M.R.1) (document ECE/TRANS/WP.29/1101/Amend.5); see https://unece.org/transport/vehicle-regulations/wp29/resolutions

ANNEX 4 PROCEDURE FOR REAR-END IMPACT TEST

1. PURPOSE AND SCOPE

1.1. The purpose of the test is to simulate the conditions of rear-end impact by another vehicle in motion. The test should make it possible to verify whether the vehicles referred to in paragraph 1 of this Regulation meet the requirements defined in paragraph 5 hereof concerning the behaviour of the structure of the passenger compartment in a rear-end collision.

2. INSTALLATIONS, PROCEDURES AND MEASURING INSTRUMENTS

2.1. Testing ground

The test area shall be large enough to accommodate the impactor (striker) propulsion system and to permit after-impact displacement of the vehicle impacted and installation of the test equipment. The part in which vehicle

impact and displacement occur shall be horizontal, flat and smooth and have a coefficient of friction of not less than 0.5.

2.2. Impactor (striker)

2.2.1. The impactor shall be of steel and of rigid construction.

2.2.2. The impacting surface shall be flat, not less than 2,500 mm wide, and 800 mm high, and its edges shall be rounded to a radius of curvature of between 40 and 50 mm. It shall be covered with plywood boards 20 ± 2 mm thick.

2.2.3. At the moment of impact the following requirements shall be met:

2.2.3.1. the impacting surface shall be vertical and perpendicular to the median longitudinal plane of the impacted vehicle;

2.2.3.2. the direction of movement of the impactor shall be substantially horizontal and parallel to the median longitudinal plane of the impacted vehicle;

2.2.3.3. the maximum lateral deviation tolerated between the median vertical line of the surface of the impactor and the median longitudinal plane of the impacted vehicle shall be 300 mm. In addition, the impacting surface shall extend over the entire width of the impacted vehicle;

2.2.3.4. the ground clearance of the lower edge of the impacting surface shall be 175 ± 25 mm.

2.3. Propulsion of the impactor

The impactor may either be secured to a carriage (moving barrier) or form part of a pendulum.

2.4. Special provisions applicable where a moving barrier is used

2.4.1. If the impactor is secured to a carriage (moving barrier) by a restraining element, the latter must be rigid and be incapable of being deformed by the impact; the carriage shall at the moment of impact be capable of moving freely and no longer be subject to the action of the propelling device.

2.4.2. The velocity of impact shall be between 35 and 38 km/h.

2.4.3. The aggregate weight (mass) of carriage and impactor shall be 1,100 ± 20 kg.

2.5. Special provisions applicable where a pendulum is used

2.5.1. The distance between the centre of the impacting surface and the axis of rotation of the pendulum shall be not less than 5 m.

2.5.2. The impactor shall be freely suspended by rigid arms rigidly secured to it. The pendulum so constituted shall be substantially incapable of being deformed by the impact.

2.5.3. Arresting gear shall be incorporated in the pendulum to prevent any secondary impact by the impactor on the test vehicle.

2.5.4. At the moment of impact the velocity of the centre of percussion of the pendulum shall be between 35 and 38 km/h.

2.5.5. The reduced mass "m_r" at the centre of percussion of the pendulum is defined as a function of the total mass "m", of the distance "a" * between the centre of percussion and the axis of rotation, and of the distance "l" between the centre of gravity and the axis of rotation, by the following equation:

$$m_r = m \cdot \frac{1}{a} \cdot$$

* It is recalled that the distance "a" is equal to the length of the synchronous pendulum of the pendulum under consideration.

2.5.6. The reduced mass "m_r" shall be 1,100 ± 20 kg.

2.6. General provisions relating to the mass and velocity of the impactor

If the test has been conducted at an impactor velocity higher than those prescribed in paragraphs 2.4.2. and 2.5.4. and/or with a mass greater than those prescribed in paragraphs 2.4.3. or 2.5.6. and the vehicle has met the requirements prescribed, the test shall be considered satisfactory.

2.7. State of vehicle under test

2.7.1. The vehicle under test shall either be fitted with all the normal components and equipment included in its unladen kerb weight or be in such condition as to fulfil this requirement so far as the components and equipment of concern to the passenger compartment and the distribution of the weight of the vehicle as a whole, in running order, are concerned.

2.7.2. The fuel tank must be filled to at least 90 % of its capacity with a liquid having a density close to that of the fuel normally used. All other systems (brake-fluid, header tanks, radiator, etc.) may be empty.

2.7.3. A gear may be engaged and the brakes may be applied.

2.7.4. If the manufacturer so requests, the following derogations shall be permitted;

2.7.4.1. The technical service responsible for conducting the test may allow the same vehicle as is used for tests prescribed by other Regulations (including tests capable of affecting its structure) to be used also for the tests prescribed by this Regulation.

2.7.4.2. The vehicle may be weighted to an extent not exceeding 10 % of its unladen kerb weight with additional weights rigidly secured to the structure in such a way as not to affect the behaviour of the structure of the passenger compartment during the test.

2.8. Measuring instruments

The instruments used to record the speed referred to in paragraph 2.4.2. and 2.5.4. above shall be accurate to within 1 %.

3. RESULTS

3.1. To measure the residual longitudinal space, the amount of longitudinal displacement of the vertical projection of the floor of the "R" point of the vehicle's rearmost seat in relation to a reference point on a non-deformed part of the vehicle structure shall be determined.

4. EQUIVALENT TEST METHODS

4.1. Equivalent test methods are permitted provided that the conditions referred to in this Regulation can be observed either entirely by means of the substitute test or by calculation from the results of the substitute test.

4.2. If a method other than that described in paragraph 2. above is used, its equivalence must be demonstrated.